Fiscal Year:	FY 2019	**Task Last Updated:**	FY 03/08/2019
PI Name:	Goukassian, David A M.D., Ph.D.		
Project Title:	Space Relevant Radiation-Induced Cardiovascular Disease Risk Thresholds: Effect of Sex on the Outcome		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI moved to Icahn School of Medicine at Mount Sinai from Temple University in October 2018.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-SRHHC. Appendix E: Space Radiobiology and Human Health Countermeasures Topics
Start Date:	05/10/2018	**End Date:**	10/17/2018
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/17/2018 due to PI move to Icahn School of Medicine at Mount Sinai from Temple University (Ed., 2/10/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kishore, Raj Ph.D. (Temple University)		
Grant/Contract No.:	80NSSC18K0921		
Performance Goal No.:			
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Task Description:	During the future Moon, near Earth asteroids, and Mars missions, astronauts will be exposed to higher total doses of space irradiation (IR) (~0.4-0.5 Gy) from galactic cosmic rays (GCR). Most of what we know about harmful effects of IR on cardiovascular (CV) system is from epidemiological studies of long-term survivors of cancer radiotherapy (RT). A recent study of 2,168 women who underwent RT for breast cancer has shown that the rates of major coronary events increased linearly with the mean dose to the heart by 7.4% per Gy, with no apparent lower or upper threshold. In this study, it was determined that average of the mean doses to the whole heart was 4.9 Gy with the range of 0.03 - 27.72 Gy. Furthermore, metabolomics studies, in patients undergoing hematopoietic stem cell (HSC) transplantation as part of cancer treatment (1.25 Gy total-body irradiated), identified seven urine-based biomarkers with distinct differences between pre- and post-exposure samples. The levels of these markers were found to be sex-dependent suggesting that separate biomarker signatures may exist for males and females. Hypotheses: Our central hypothesis is that low-dose proton and HZE (high energy) particle IR-induced biological responses are long-lasting, IR type- and dose-dependent and may augment excess relative risk (ERR) estimates for the development of CV diseases during and after long-duration space missions. In addition, we hypothesize that sex differences could further modify radio-biologically effective (RBE) IR thresholds for CV risk estimates. Gene expression and epigenetic modifications in protein and microRNA (miRNA) in exosomes from the blood (e.g., plasma/serum) may be altered before the onset of the cardiac symptoms, which could be used as potential biomarkers to predict the CV disease risks. We will test our hypotheses with the following specific aims: AIM 1. Determine the longitudinal effect of IR type, dose, and sex on cardiovascular physiology in wild type mice and ApoE null mice after full-body 5-ion simplified mixed field and gamma radiation. AIM 2. Determine space-type IR mediated modulations in exosomal cargo in the blood, and determine whether these changes are associated with alterations in the heart function, structure and vasculature before manifestation of clinical symptoms. AIM 3. Utilize known and newly identified bio-markers in the blood to develop human-relevant point-of-care tests (POCT) for predicting and monitoring possible CV alterations before and during the space flights. We anticipate that the results of our proposed work may be beneficial for human space exploration and could (1) Determine single, low-dose 1H, 56Fe, and mixed field dose-responses, radio-biologically effective IR thresholds in the heart and cardiac vasculature, and whether sex differences could modify radio-biologically effective IR thresholds for CV risk estimates; 2) Determine whether space radiation leads to modifications in the circulating exosomal cargo contents and whether IR-induced exosomal cargo modulations are reflective of subclinical changes in the cells and organs of origin; 3) Ascertain if modulations of exosomal cargo may be representative of chronic oxidative stress and inflammation and could serve as early bio-markers of IR-induced CV disease initiation and progression; 4) Integrate physiological CV endpoint data sets with gene expression and epigenetic data to identify bio-markers in bio-fluids that could be used for prediction of asymptomatic CV disease in the setting of space IR, which will include known early and intermediate bio-markers of cardiac damage, inflammation, and oxidative stress, as well as currently unknown novel radiation-associated cardiac bio-markers. NOTE (Ed., 2/10/2020): New grant "Space Relevant Radiation-Induced Cardiovascular Disease Risk Thresholds: Effect of Sex on the Outcome--80NSSC19K1079" established at Icahn School of Medicine at Mount Sinai due to PI move there from Temple University.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We anticipate that the results of our work could be beneficial for human space exploration as well as for the Earth-based applications on several levels -- (1) determine whether low dose space-type and terrestrial IR may present an increased risks for CV disease development during and after prolonged space missions, as well as after conventional and particle cancer radiotherapy; (2) determine the underlying molecular signaling of CV alterations; (3) identify bio-markers in the blood that could be used for prediction of asymptomatic CV disease, which will include known early and intermediate bio-markers of cardiac damage, as well as currently unknown novel cardiac biomarkers; (4) the identification of sub-clinical CV disease biomarkers that could be used for monitoring the effectiveness of mitigating factors for prevention and treatment of IR-induced CV diseases in space and in Earth-bound civilian population, in general.
	In late September 2018, we irradiated at the NASA Space Radiation Laboratory (NSRL) and the Brookhaven National Laboratory (BNL) gamma facility 660 ApoE null male mice and 480 C57Bl/6J male mice, first two cohorts of our longitudinal life time studies. This longitudinal life-time study is designed to have 6 harvesting time points (14 days and 1, 6, 12, 18 months, and one end of life time point – 20 or 24 months) and the number of animals in all cohorts will be 15 for each dose/time point. We have already performed cardiac function analyses and harvested tissues and blood for the first two early time points, i.e., 14 days and 1 month. We report here our findings for these two early time points. The next 6-months after initial radiation exposure time point is scheduled for late March-early April, 2019. In terms of selection of radiation parameters, we utilized the following doses, energies, and ions, as recommended by Radiation Biology Element Management: (i) Gamma IR - 1, 2, and 4 Gy, using ApoE null MALE mice for all time points indicated above. (ii) 5-Ion Simplified Mixed field IR - 0.5, 0.75, and 1.5 Gy, adjusted to 500 MeV/n, using ApoE null MALE mice for all time points indicated above. (iii) Gamma IR - 2 and 4 Gy, using WT-C57BL/6J MALE mice for all time points indicated in section above. (iv) 5-Ion Simplified Mixed field IR - 0.75 and 1.5 Gy, adjusted to 500 MeV/n, using WT-C57BL/6J MALE mice for all time points indicated above. Cardiac function was assessed non-invasively in all control and irradiated ApoE null and C57Bl/6J mice by transthoracic echocardiography (ECHO) at baseline (before IR) and then 14 and 28 days after irradiation. • Compared to control non-irradiated mice of either genotype or radiation type, the ECHO measurements revealed substantial decreases in the global systolic function, indicating dysfunction in heart contractile function. • Decreases in the global systolic function of the heart detected by ECHO were associated with mild to moderate

Task Progress:	decreases in left ventricular posterior wall thickness in SYSTOLE but not diastole, indicating that decreases in fractional shortening percent are starting to affect cardiac morphology as early as day 14 and continuing to decrease on day 28 after irradiation. Cardiac morphology. To evaluate whether decrease in global systolic function of the heart may have already affected the cardiac morphology, multiple cross sections of the heart from all ApoE null and C57Bl/6J mice, for 14 and 28-day time points, were stained with H&E are examined microscopically by clinical pathologist blindfolded to the treatment conditions at the Comparative Pathology Laboratory of the Center for Comparative Medicine and Surgery at Mount Sinai. • Majority of the samples on day 14 after gamma or mix ion radiation did not reveal significant identifiable microscopic lesions. • However, by 28 days after radiation appreciable histological chances showed significant multifocal myofiber disarray with mildly increased variation in myofiber diameter and swollen myocytes. • These changes were more prominent in 1 and 2 Gy gamma-irradiated mice of both ApoE null and C57Bl/6J mice. To summarize our findings for 14 and 28-day time points, we found that: • Left Ventricular (LV) Fractional Shortening Percent is Decreased After all Doses of Gamma and Mix Ion Radiation in both ApoE null and C57Bl/6J mice; • LV Posterior Wall Thickness (LV PWth) is Decreased in Systole in only in Gamma Irradiated C57Bl/6J mice; • For ApoE null mice there is trend of decreased LV PWth in Systole after both Gamma and Mix Ion Radiation in both ApoE null and C57Bl/6J mice; • LV PWth in Diastole is NOT affected after both Gamma and Mix Ion Radiation in neither ApoE null, nor C57Bl/6J mice; • Histological analyses of H&E stained slides at 28 days after radiation showed – (i) multifocal myofiber disarray with mildly increased variation in myofiber diameter; (ii) swollen myocytes; (iii) mild to moderate sarcoplasmic swelling; (iv) minimal multifocal mineralization of few myofibers within papillary muscles. Preliminary interpretations: Both, Gamma and Mix Field Ion full body radiation, at doses 1-4 Gy for gamma and 0.5-1.5 Gy for Mix Ions, decreased global systolic function of the heart in both ApoE null and C57Bl/6J mice at 14 and 28 days after radiation. Changes in diastolic function of the heart were detected at 14 or 28-day time point after any of the IR doses in both ApoE null or C57Bl/6J. Presence of global Systolic dysfunction already by 14 and 28 days measured by transthoracic echocardiography and identification of the first detectable changes in the heart tissue only by day 28 after irradiations may indicate that changes in cardiac function (contractility but not relaxation at this time) could be detected in advance of morphological changes in the heart tissue, suggesting that echocardiography may be one of the early tests to identify cardiovascular disease risk development. NOTE (Ed., 2/10/2020): Grant continues as "Space Relevant Radiation-Induced Cardiovascular Disease Risk Thresholds: Effect of Sex on the Outcome--80NSSC19K1079" due to PI move to Icahn School of Medicine at Mount Sinai from Temple University. See that project for future reporting.
Bibliography Type:	Description: (Last Updated: 04/23/2026)